

ACM ICAIF 2026 Trading Agent Competition

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1 INTRODUCTION

Trading in equity markets involves a continuous stream of decisions: how much capital to allocate, when to rebalance, and how much risk to bear as market conditions and portfolio states evolve. Under noisy signals, transaction costs, portfolio constraints, and shifting market regimes, trading performance depends largely on the quality of these repeated decisions.

Sequential decision-making of this kind lies at the heart of agentic systems, making trading a natural application for autonomous agents that map market observations directly to portfolio actions [4–6, 8, 9]. In practice, such agents can be built in many ways, drawing on expert rules, reinforcement learning, return forecasting, large language models, or combinations thereof, and new designs continue to emerge.

However, these agents are often developed and tested in isolation, each on its own data sources, market assumptions, and simulation environments, making results across studies difficult to compare or reproduce. Recent benchmarks [2, 3, 7] have taken steps towards standardised evaluation, but they (1) are tailored to specific agent families rather than open to arbitrary designs; (2) rely on static historical backtests, where evaluation data may no longer be genuinely unseen; and (3) often simplify away transaction costs, portfolio constraints, and downside risk that shape real trading outcomes.

Our competition aims to fill this gap by hosting a live trading agent arena, where agents of any design compete in a common market environment on real, post-submission data. Agents are evaluated on both profitability and risk management under realistic trading conditions. This initiative seeks to offer a common ground on which diverse trading agents can be compared fairly, and to encourage the community to explore a wider variety of agent designs for real-world financial markets.

2 TASK

The competition challenges participants to build a trading agent that manages a portfolio over a universe of $n = 30$ U.S.-listed equities. On each trading day t , the agent receives the latest market information together with its own portfolio state, and returns target portfolio weights $\mathbf{w}_{t+1}$ for the next trading day. Trading is subject to realistic constraints, including long-only positions, per-asset weight caps, and transaction costs, which are detailed in Section 3.

Three types of information are visible to the agent each day t . Market features X_t summarise recent price and volume behaviour, such as rolling returns, momentum, and volatility. Fundamental features Z_t provide company-level ratios on profitability, growth, and leverage. Daily news N_t offers an unstructured view of the market, carrying information on corporate events and sentiment that is not fully captured by numerical features. To ensure fairness and reduce data engineering overhead, all participants receive the same processed data streams in a unified observation format.

Formally, at each trading day t , the agent observes $\mathbf{O}_t = (X_t, Z_t, N_t)$ together with its portfolio state S_t , covering current weights, cash ratio, and drawdown. Each agent π_θ implements the mapping:

$$\pi_\theta : (\mathbf{O}_t, S_t) \rightarrow \mathbf{w}_{t+1} \in \mathcal{W},$$

where $\mathbf{w}_{t+1}$ is the vector of target portfolio weights executed on day $t + 1$, and $\mathcal{W}$ is the feasible set defined by the trading constraints. The task is inherently sequential: each action affects the portfolio state through execution and costs, and returns compound over the evaluation horizon. There is no restriction on how π_θ is built, as long as it implements the above mapping.

To support agent development, we provide a historical dataset over a separate pool of public equities with no overlap with the evaluation universe, together with an environment SDK that replicates the observation format, action format, and trading rules of the official evaluation. Before the deadline, participants can submit their agents multiple times to an online validation environment, which runs on a held-out historical period, verifies that the agent operates correctly under the official protocol, and returns metric scores as feedback.

For final evaluation, each team submits a single containerized agent before the deadline, after which no further changes are allowed. Submitted agents are then run forward for around two weeks of real trading days, receiving each day’s actual market data and news as input. Final rankings are based solely on the evaluation protocol described in Section 3.

To accelerate participation, we provide a comprehensive starter kit with an illustrative notebook covering (1) environment setup and data access, (2) baseline agents built in several different ways, and (3) the local evaluation pipeline. This resource helps participants quickly understand the competition framework and start experimenting effectively.

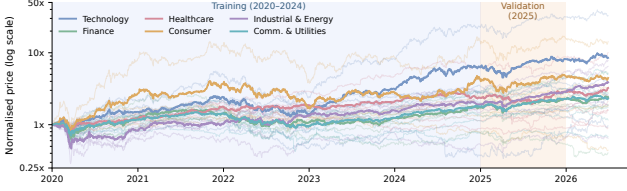


Figure 1: Normalised prices of the 30-equity universe from January 2020 to June 2026, with bold lines showing equal-weighted sector indices. Shaded regions mark the chronological data splits; the live evaluation takes place after the submission deadline.

Impact. A shared environment with identical data, rules, and costs makes studies of trading agents easier to relate and reproduce, and the post-submission evaluation offers a simple way to assess agents on data that no participant or model has seen. Beyond research, a better understanding of how different agents behave under realistic costs and risk constraints supports the development of more robust and auditable automated trading systems.

Originality. While prior competitions and benchmarks have explored financial forecasting and trading tasks [2–4, 7], to our knowledge, no existing ICAIF competition has evaluated freely designed trading agents on live, post-submission market data. Specifically, our competition offers (1) a unified market environment open to agents of any design, (2) a live evaluation phase that avoids look-ahead bias and data contamination by construction, and (3) a comprehensive assessment covering profitability, risk-adjusted performance, and risk management.

Relevance to the ICAIF community. For the ICAIF community, this competition connects research on trading agents with the practical conditions under which they would operate. It raises questions such as: *How do agents built in different ways behave under the same market conditions? Can unstructured news signals be turned into reliable portfolio decisions? Which designs remain effective once costs and risk constraints are taken into account?* Exploring these questions contributes to more robust trading systems and a broader design space for financial agents.

3 EVALUATION PROTOCOL

The performance of participants’ trading agents will be assessed under a common market environment, with identical data, trading rules, and evaluation metrics for all teams. This section describes the competition dataset, the trading rules and evaluation procedure, and the metrics that determine the final rankings.

3.1 Competition Dataset

The competition is built on a fixed universe of 30 U.S.-listed equities spanning six sector groups, covering technology, finance, healthcare, consumer, industrial and energy, and communication and utilities. The universe is chosen to cover a broad range of market behaviours and cross-sectional structures. For each asset i and trading day t , three types of data are provided:

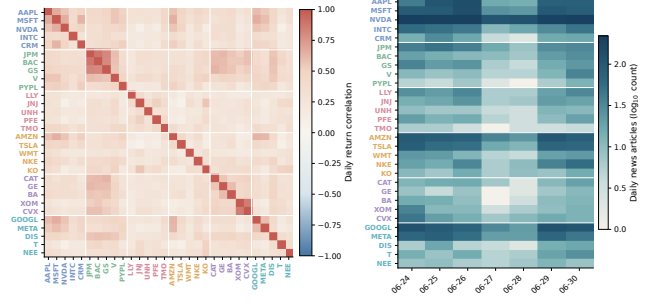


Figure 2: Cross-sectional structure of the universe: pairwise daily return correlations over 2020–2026 (left) and daily news article counts per asset over a representative week (right).

- **Market Data.** Daily open, high, low, close, and volume records under a consistent adjusted-price convention. Daily log-returns are calculated as

$$r_{i,t} = \log \left(\frac{C_{i,t}}{C_{i,t-1}} \right),$$

where $C_{i,t}$ denotes the adjusted closing price of asset i on day t .

- **Fundamental Data.** Quarterly financial statements summarised into company-level ratios on profitability, growth, and leverage.
- **News Data.** A daily stream of financial news associated with the universe, collected from public sources, deduplicated, and aligned to assets by the organisers. All participants receive the same processed news stream.

Data Splits. The dataset is organised chronologically into three parts. The *training data*, covering January 2020 to December 2024, contain market and fundamental data and are released to all participants for free use in development. The *validation environment* runs on 2025 data and mirrors the official protocol; agents can be submitted to it multiple times before the deadline, while the resulting scores do not count towards the final rankings. The *live evaluation data* consist of real market data and news arriving after the submission deadline, and are the only data used for the final rankings. In this way, agents are always evaluated on genuinely unseen data.

Exploratory Data Analysis. The dataset reflects diverse market conditions, including the sharp sell-off in early 2020, the 2021 rally, the 2022 downturn, and recovery phases in 2023–2024 (Figure 1). Figure 1 also marks the chronological data splits, with the live evaluation window lying beyond the observed data. Cross-sectional correlations are markedly stronger within sectors than across them (Figure 2, left), reflecting a rich sector structure. Meanwhile, every asset in the universe receives daily news coverage (Figure 2, right), providing a solid basis for agents that exploit unstructured information.

3.2 Trading Rules and Evaluation Procedure

While agents are free in how they make decisions, all portfolios are managed under the same rules:

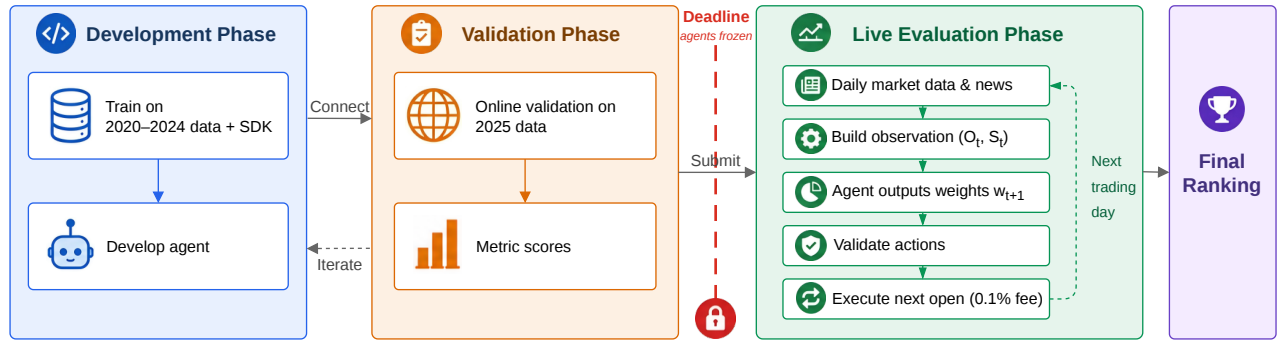


Figure 3: Overview of the competition workflow: development on historical training data, validation through the online environment, and live evaluation on post-submission market data with a daily observe–decide–execute cycle.

- **Portfolio Constraints.** Each agent starts with \$1,000,000 in cash and manages a long-only portfolio, with a maximum weight of 30% per asset and a gross exposure of at most 100%; any unallocated amount remains as cash.
- **Execution and Costs.** Target weights submitted after the close of day t are executed at the open of day $t + 1$, with a 0.1% transaction fee charged on both buy and sell notional; fractional shares are allowed. This observe-then-execute timing ensures that no decision can use same-day information.
- **Action Validation.** Invalid actions are automatically repaired before execution: negative or oversized weights are clipped, excessive gross exposure is scaled down, and if an agent crashes or returns no action, its previous target weights are retained. The episode always continues after repair, while every violation is recorded and counted in the evaluation.

Live Evaluation Procedure. After the submission deadline, all agents are frozen and run forward for around two weeks of real trading days (Figure 3). On each day, the organisers’ data pipeline collects the actual market data and news, builds the observation in the official format, queries each agent for its target weights, and executes them on the next trading day under the rules above. All agents therefore experience exactly the same sequence of market conditions, and their full decision histories are logged for verification and analysis.

3.3 Evaluation Metrics

Agents will be assessed across four key dimensions: profitability (M1–M2), risk-adjusted performance (M3–M4), risk management (M5–M7), and execution quality (M8–M9).

Profitability. These metrics measure the wealth generated over the evaluation horizon:

- **M1: Cumulative Return** measures the total growth of portfolio value over the evaluation period.
- **M2: Daily Win Rate** measures the fraction of trading days with positive portfolio returns, reflecting the consistency of profits.

Risk-adjusted Performance. Raw returns can be inflated by taking excessive risk, so returns are also assessed relative to the volatility incurred:

- **M3: Sharpe Ratio** captures return per unit of volatility.
- **M4: Sortino Ratio** refines the Sharpe ratio by penalising only downside volatility.

Risk Assessment. Equity portfolios remain exposed to drawdowns and tail events, so downside risks are evaluated directly [1]:

- **M5: Maximum Drawdown (MDD)** measures the worst peak-to-trough decline, critical for risk-averse investors.
- **M6: Value at Risk (VaR)** estimates the maximum expected loss over a given horizon at a specific confidence level.
- **M7: Expected Shortfall (ES)** calculates the average loss exceeding the VaR threshold, offering a more comprehensive view of tail risk.

Execution Quality. Frequent or careless trading erodes returns and signals fragile decision-making, so trading behaviour itself is also evaluated:

- **M8: Turnover and Cost Rate** measure the total traded notional and transaction fees relative to portfolio value.
- **M9: Violation Rate** counts the fraction of trading days on which an agent’s action required repair.

Scoring Procedure and Winner Selection. Each metric is computed separately over the live evaluation period. For consistency, all metrics are transformed so that lower scores indicate better performance; higher profitability translates into lower scores for return-based metrics. Final rankings are determined by averaging an agent’s ranks across all metrics, and the submission with the *lowest average rank* is declared the winner. Ties are resolved based on the time of submission. This evaluation favours agents that combine profitability, disciplined risk management, and reliable execution.

4 LOGISTICS

Competition Timeline. The competition will follow a structured timeline:

- **Preparation Phase (14 Sep–4 Oct):** We will finalise the dataset and data pipeline, build the competition platform with the validation and live evaluation environments, and prepare the starter kit. Concurrently, we will promote the event via social media, academic mailing lists, and a Discord channel.

- **Competition Phase** (5 Oct–23 Oct): Participants may register individually or as teams and will receive the training data, the environment SDK, and the starter kit. Agents can be submitted to the online validation environment throughout this phase. Final agents are due by 23 Oct, after which no further changes are allowed.
- **Live Evaluation Phase** (26 Oct–6 Nov): All submitted agents will be run forward on real daily market data and news for two weeks of trading days, as described in Section 3.
- **Final Result & Presentation Phase** (9 Nov–17 Nov): The final results will be announced on 9 Nov, and the top three teams will be invited to present their solutions at ICAIF 2026.

Organizers. The competition is organized by a diverse team of researchers and practitioners from the National University of Singapore (NUS), University College London (UCL), and the University of Edinburgh, combining expertise in data science, machine learning, finance, and competition organization:

- **Prof. Anthony K. H. Tung (NUS):** A renowned expert in data mining with over 20 years of experience and a strong record of leading large-scale research initiatives.
- **Prof. Hao Ni (UCL):** A world-leading expert in rough path theory and machine learning, with experience in co-leading major research programs and organizing high-impact ICAIF competitions in 2023 and 2024.
- **Prof. Lukasz Szpruch (University of Edinburgh):** A leading figure in AI and finance, serving as Programme Director at the Alan Turing Institute and advancing responsible AI in financial applications.
- **Dr. Yihao Ang (NUS):** A Research Fellow specializing in time-series data mining and former team lead of the ICAIF 2024 Data Competition champion team. He oversees competition design and technical implementation.
- **Qiang Wang (NUS):** Brings industry insights from his tenure as Tech Lead in several technology firms, contributing to data collection and problem formulation.
- **Yifan Bao (NUS):** A Ph.D. student focused on time-series forecasting and generation, instrumental in maintaining the competition’s online resources and shaping the evaluation protocol.
- **Xinyu Xi (NUS):** A Research Assistant with experience organizing impactful AI competitions, supporting technical infrastructure and participant engagement.

Together, the team brings academic rigor, industry relevance, international collaboration, and diversity across genders and career stages, creating a strong foundation for a successful and impactful competition.

Participant Recruitment. Our recruitment strategy leverages the extensive network of the National University of Singapore, particularly through its AI Institute (NUSAI), to reach a broad and diverse pool of participants. Additionally, the international backgrounds and affiliations of the organizing team will enable promotion through multiple institutions and networks, ensuring global visibility and engagement.

A dedicated competition website will serve as the central hub for information and as the hosting platform. To support participants of all experience levels, we will provide supplementary learning

materials and a comprehensive starter kit during the registration period, ensuring the competition remains inclusive and educational. Moreover, top-performing teams will be invited to present their work at a follow-up workshop, offering opportunities to engage with leading researchers and explore future collaborations.

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